Embedded System

Computer

Define ?

⇒ combination of H/W & S/W

H/W → CPU/MCU

S/W → program

H/W

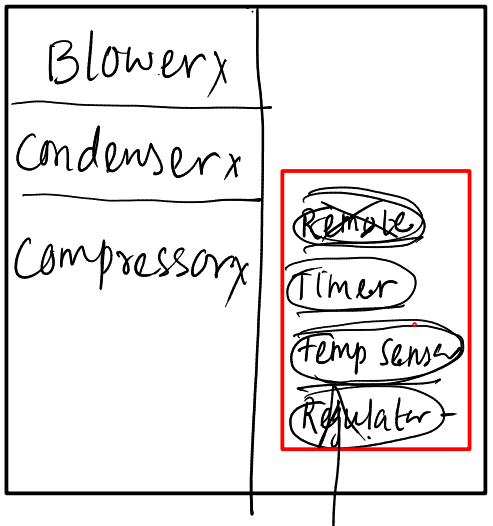
S/W

Emb System is generally used for specific task

↳ Home Appliance

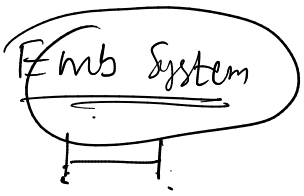
Real time OS / Mission Critical

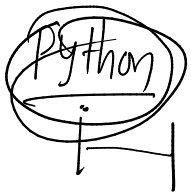
Air Conditioner ← larger System



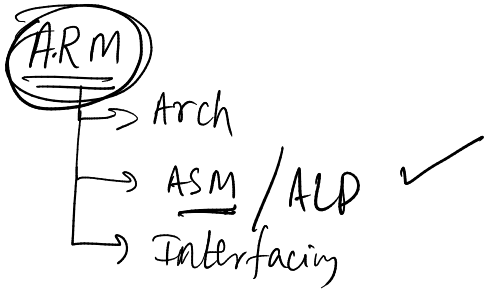
Computer based sub-system, Integrated in a larger System

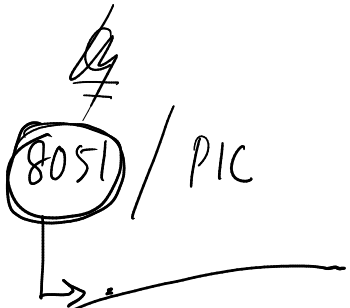
DSP, Matlab, DD, VHDL, IoT,











C is not MUST for emb-system

ALP is MUST for Emb. System.

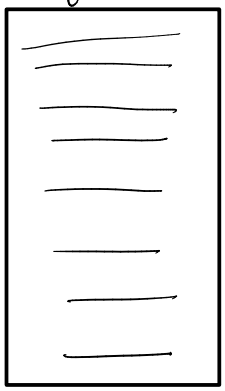
ARM - 4 weeks
Busp - 4 "
8051 - 2 "

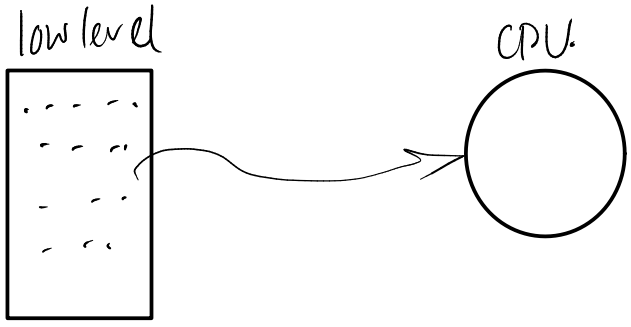


12 weeks
4 weeks

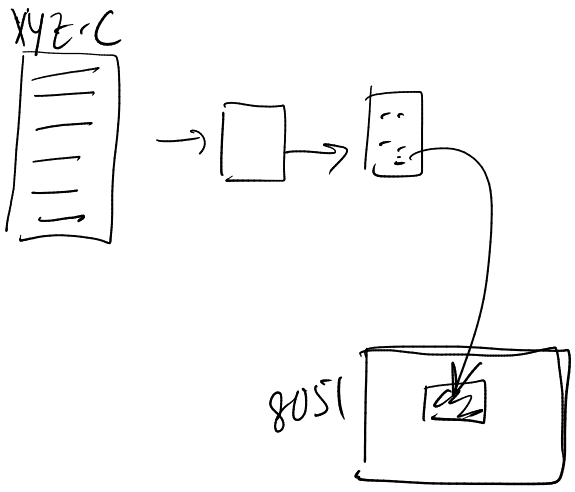
C prog ~~Faster~~ than ALP

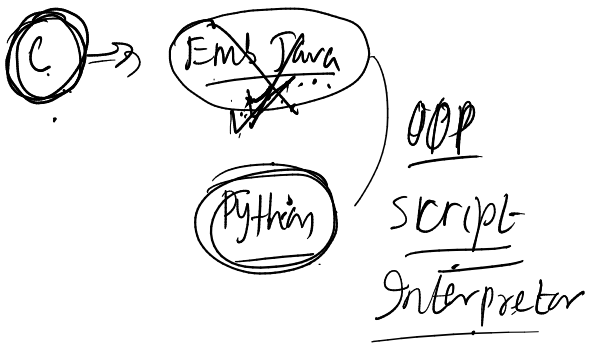
Source Code
Highlevel

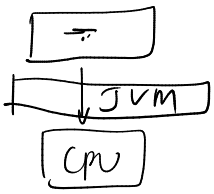




C / Java / .Net / Python / CPP



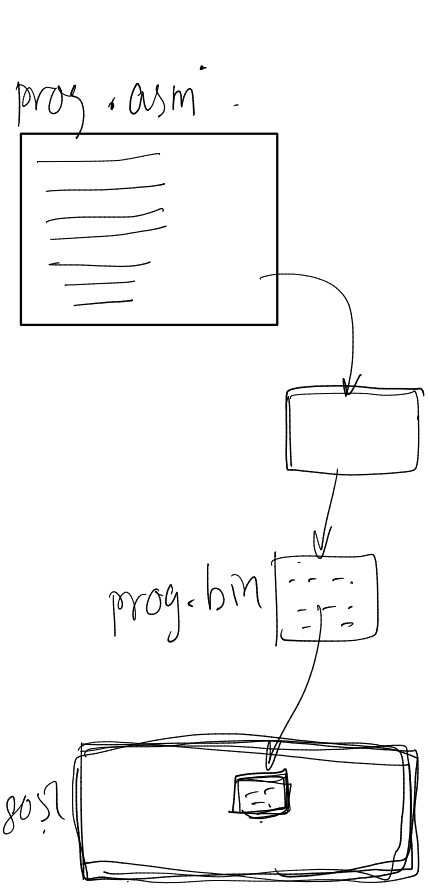


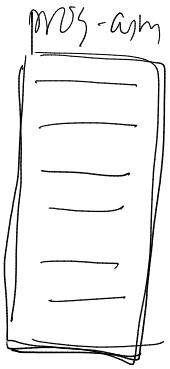


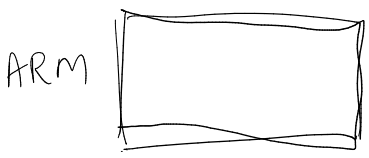
ALP vs C

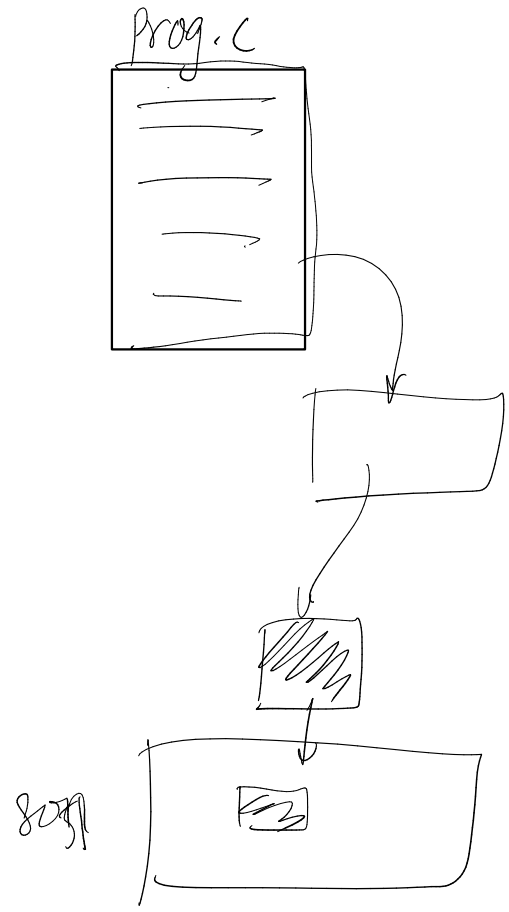
- ✓ Easy
- ✓ Portable
- ✓ Debug ✓
- ✓ Reusable ✓

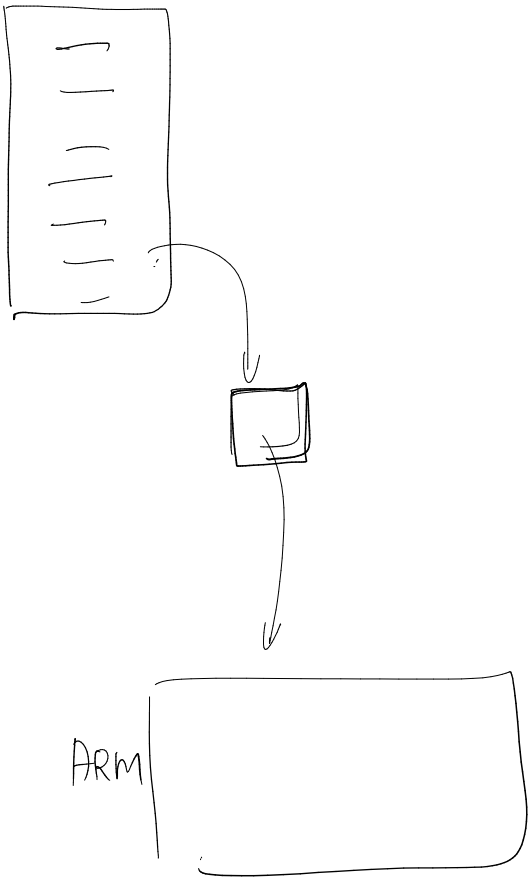
C vs Python/Java/—











function Pointer

volatile

enum vs #define

struct vs union